

# E-Commerce Platform System Requirements & Specification

Version 1 — Faithful Extraction from Source

**What this document is.** A faithful, de-duplicated reorganisation of the supplied specification. **Nothing has been invented.** Items tagged **(spec)** are stated in your source; **(inferred)** are reasonable models to confirm; **■ GAP / ■ CONFLICT** are undefined, contradictory, or ambiguous and require a decision before build.

Prepared for poster presentation / thesis project · build brief for Claude Code

# How to read this document

Three confidence levels are used throughout so accuracy is never ambiguous:

**(spec)** — stated explicitly in the source material.

**(inferred)** — implied by the source; a reasonable default that should be confirmed before coding.

■ **GAP** / ■ **CONFLICT** — named-but-undefined, internally contradictory, or ambiguous. Listed in full in §10.

Do not begin coding before the “Decisions needed” section (§11) is answered. Every unresolved item is a point where the running software will silently diverge from what the poster claims.

## 1. System overview

A fashion / clothing **e-commerce platform** for the Bangladesh market, with two faces:

- **Customer storefront** — browse, search, cart, wishlist, checkout, order tracking, profile.
- **Admin / Manager back-office** — catalogue, orders, customers, analytics, CMS, staff management.

Distinctive design decisions taken directly from the source:

- **Manual bKash ‘Send Money’ payment** — no payment-gateway API; customer submits a sender mobile number (transaction ID and screenshot are *optional*).
- **Manual fulfilment** — a manager enters a **Parcel ID** by hand, then clicks *Confirm*.
- **Email OTP verification** (placement ambiguous — see §7).
- **Bangladesh address model** — Division / District / Upazila / Union / Postal code.

## 2. Actors & role-based access control

Three actors (spec): **Customer**, **Manager**, **Super Admin**.

| Capability                                             | Customer | Manager              | Super Admin |
|--------------------------------------------------------|----------|----------------------|-------------|
| Browse / cart / wishlist / checkout / track own orders | Y        | —                    | —           |
| Manage own profile                                     | Y        | Y                    | Y           |
| Manage products (add / edit / delete / stock)          | —        | Y                    | Y           |
| Manage orders (confirm / cancel / status / courier)    | —        | Y                    | Y           |
| Manage inventory                                       | —        | Y                    | Y           |
| View analytics                                         | —        | Y                    | Y           |
| View customers                                         | —        | Y                    | Y           |
| Website management / CMS                               | —        | partial <sup>1</sup> | Y           |
| Add / edit / delete categories                         | —        | no <sup>2</sup>      | Y           |
| Manage managers (add / remove / reset pwd)             | —        | —                    | Y           |
| Store settings                                         | —        | —                    | Y           |

<sup>1</sup> Source lets a manager edit only **Featured Products** in Website Management — unclear if managers get the full CMS.  
**Decision needed.**

<sup>2</sup> Source: “**Only Admin** can Add/Edit/Delete Category” — yet managers manage products, which depend on categories.  
**Confirm.**

### 3. Admin / back-office modules

#### Defined in source (spec):

- **Dashboard** — cards: Total / Today's / Monthly Revenue; Total / Pending / Delivered / Cancelled Orders; Total Customers; Total Products. New orders fire a dashboard notification.
- **Products** — Add / Edit / Delete; Images; Variants; Stock; Price; Discount; Featured; Status. Fields: Name, Description, Category, Brand, Price, Discount Price, Stock, SKU, Images, Sizes, Colors, Material, Weight, Tags, Status.
- **Categories** — Men, Women, Kids, Accessories (→ Cap → Mask), Shoes, Winter, Summer. Admin-only CRUD.
- **Orders** — filtered by Pending / Confirmed / Cancelled. Per order: Order ID, Customer ID, products, quantity, price, shipping address, Parcel ID, order-placed date. Manager enters Parcel ID *before* Confirm; OTP email verification required; customer address + mobile captured at checkout and shown to staff.
- **Customers** — Name, Email, Phone, Address, Total Orders, Total Spending. **View-only** for admin + manager.
- **Managers** (Super-Admin only) — Add / Edit / Remove; assign email + password; Reset Password. Panel scope: Dashboard, Products, Orders, Customers, Inventory, Analytics.
- **Sales & Analytics** — Revenue, Profit, Orders, Products Sold, Top Products, Top Categories, Customer Growth, Monthly / Weekly / Daily Sales. Charts: Bar, Pie, Line.
- **Website Management (CMS)** — Homepage Banner, Hero Slider, Featured Products, Latest Collection, About Us, Contact Information, Social Links, Footer.
- **Profile** — personal info; contact (primary + secondary email, mobile 1–3, address with required Line 1 / City / Upazila / Division / Postal); account settings; actions.

#### ■ Named in the module list but NOT specified (gaps to define):

- **Inventory** — distinct from product *stock*? Stock-in/out, low-stock alerts?
- **Coupons & Discounts** — yet the cart has a 'Coupon Code' field. Rules undefined.
- **Reviews** — storefront reviews exist; admin moderation flow undefined.
- **Notifications** — channels and triggers undefined.
- **Activity Logs, Reports, Settings** — contents undefined.

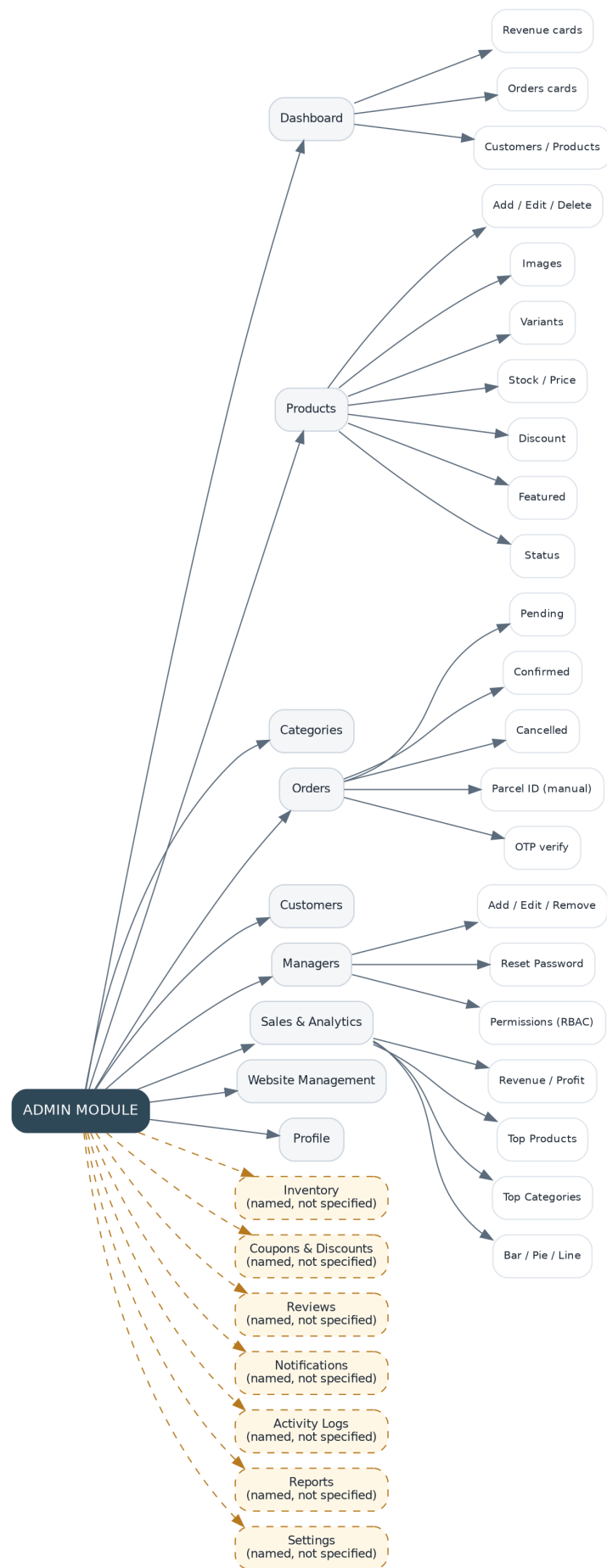


Fig 1. Admin module map. Solid = specified; dashed amber = named but undefined.

## 4. Customer storefront modules (spec)

- **Home** — Announcement Bar, Nav, Hero Banner, Featured Categories, New Arrivals, Best Sellers, Flash Sale, Trending, Featured Collection, Why Choose Us, Reviews, Instagram Gallery, Newsletter, Footer.
- **Shop** — filters (Category, Brand, Price, Size, Color, Discount, Availability), Sort, Grid, Pagination, Search.
- **Categories** — Men, Women, Kids, Shoes, Accessories, Winter Collection, Summer Collection, Sale. ■ *differs from admin list (§10).*
- **Product Details** — Gallery + Zoom, Name, Brand, Price, Discount, Rating, Stock, Size + Color, Quantity, Add to Cart, Buy Now, Wishlist, Share, Description, Specifications, Reviews, Related, Recently Viewed.
- **Cart** — Items, Qty, Remove, **Coupon Code**, Shipping Charge, Summary, Checkout.
- **Checkout** — Customer Info, Shipping Address, Billing Address (*optional*), Shipping Method, Payment Method, Summary, Place Order.
- **Payment** — bKash instructions, store number, sender mobile, Transaction ID (*optional*), Screenshot (*optional*), Submit.
- **Order Success / Tracking / My Orders** — Order ID, Payment Status, Estimated Delivery; Parcel ID, Courier, Timeline; Current / Completed / Cancelled, Invoice.
- **Customer Profile** — Dashboard, Personal Info, Address Book, Wishlist, My Orders, Payment History, Notifications, Change Password / Email.
- **Authentication** — Login, Register, Verify Email (OTP), Forgot / Reset Password.
- **Static** — About, Contact (+ Map), FAQ, Privacy, Terms, Return & Refund, Shipping Policy, Footer.

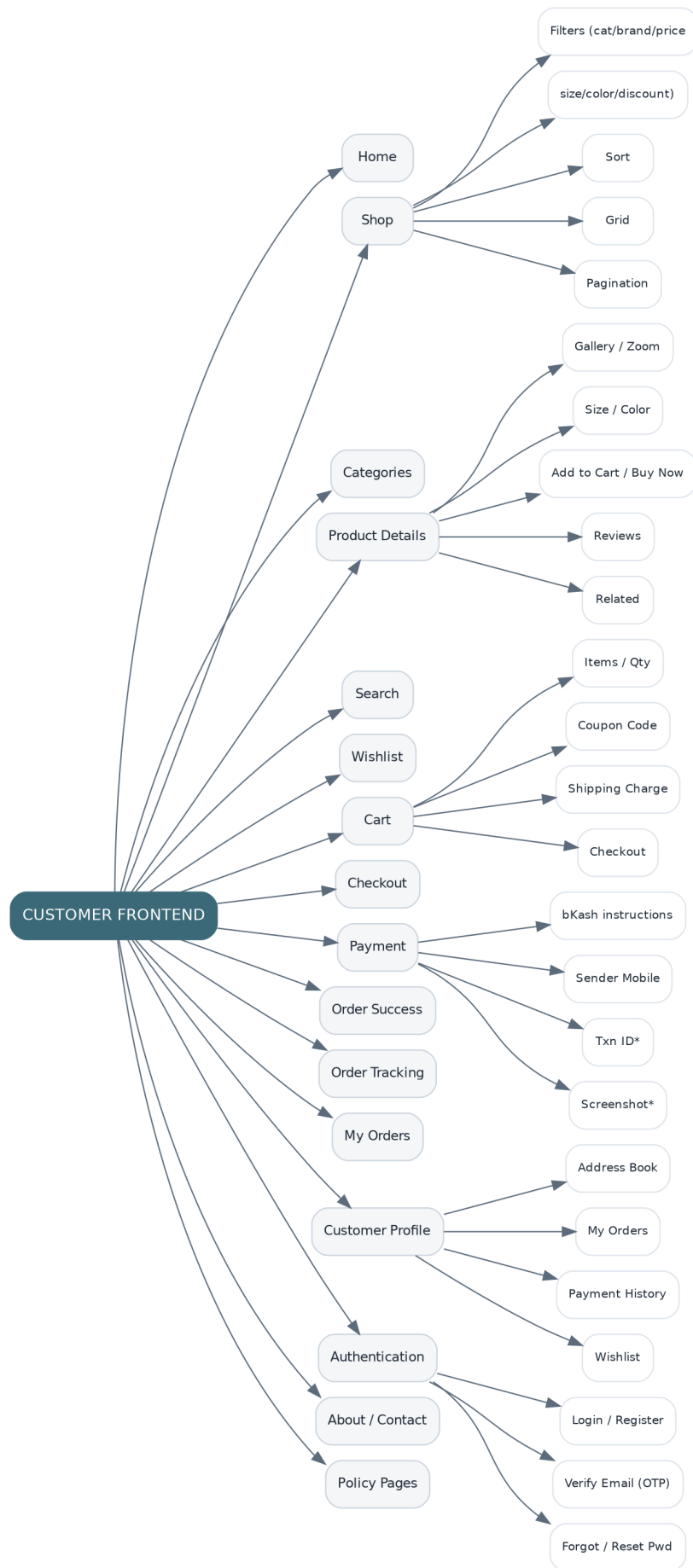


Fig 2. Customer storefront module map.

## 5. Data model

Entities and relationships are **inferred** from the field lists in the source — confirm before generating migrations. Amber entities map to undefined modules; their fields are provisional.

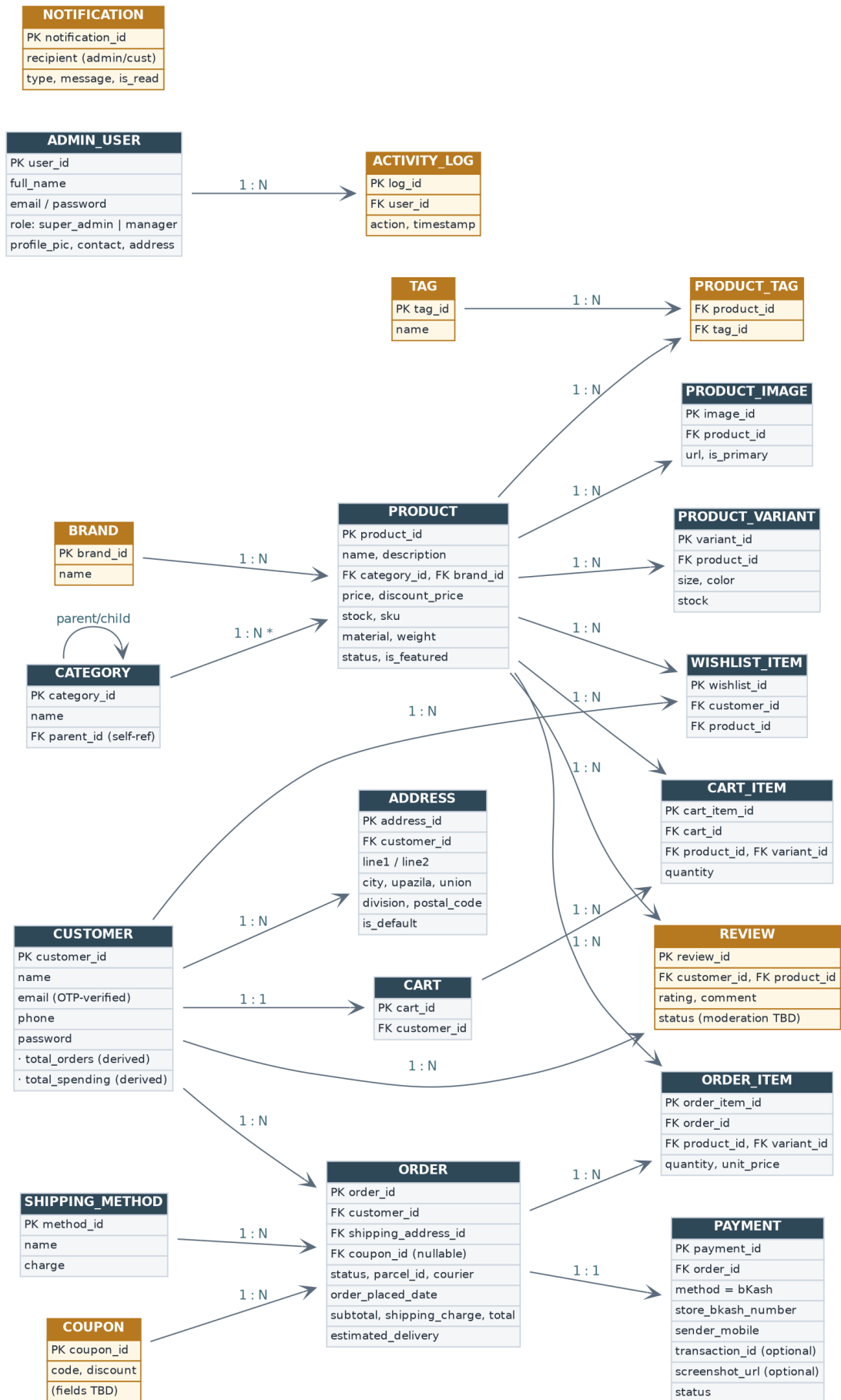


Fig 3. Entity relationship model (inferred). Amber = provisional / undefined module.



### Modelling decisions to confirm:

- **Product→Category is 1:N** per the single 'Category' field — most stores use M:N.
- **Sizes/Colors as variants** — modelled as PRODUCT\_VARIANT rows so each size+color carries its own stock; the source lists them as plain fields.
- **Brand** promoted to its own table because a Brand filter exists.

## 6. Order lifecycle (state machine)

The source lists **only three** states (Pending, Confirmed, Cancelled), but the dashboard, Order Tracking and My Orders imply more. Dashed states are **implied, not explicit** — confirm the full lifecycle or Claude Code will guess.

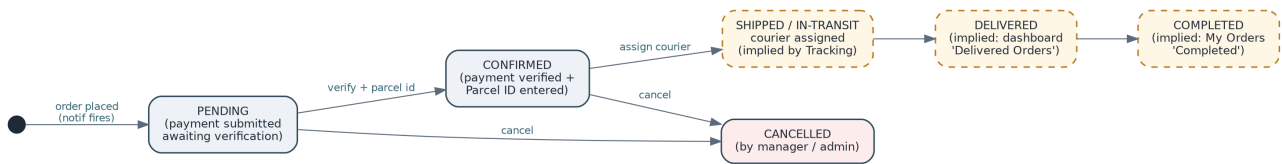


Fig 4. Order lifecycle. Solid = explicit in source; dashed = implied by other screens.

## 7. Customer purchase flow

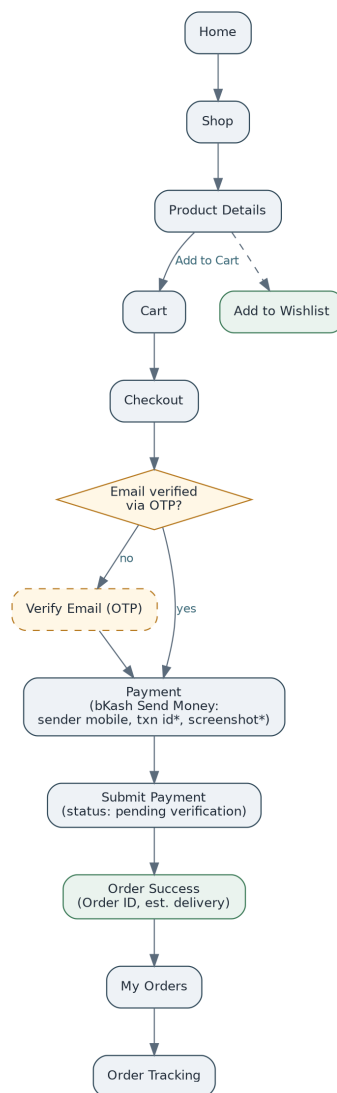


Fig 5. Customer purchase flow with OTP and manual-bKash branch.

■ **OTP placement conflict.** Authentication lists 'Verify Email (OTP)' at **registration**, while Orders says email must be verified at **order time**. Decide: register-time, checkout-time, or both.

## 8. Payment & fulfilment model (the distinctive part)

1. Customer manually sends money to the store's bKash number.
2. Customer submits **sender mobile number**; transaction ID and screenshot are **optional**.
3. Order is created in **Pending**; dashboard notification fires.
4. A manager manually **verifies** the payment, enters the **Parcel ID**, clicks **Confirm**.
5. Courier assigned; customer tracks via Parcel ID + timeline.

■ **Verification weakness to address (and defend on your poster).** Because txn ID and screenshot are optional, a Pending order may carry **no verifiable proof of payment**. Either require at least one proof, or be ready to justify the friction-vs-fraud trade-off — examiners will ask.

## 9. Reference architecture (stack to be decided)

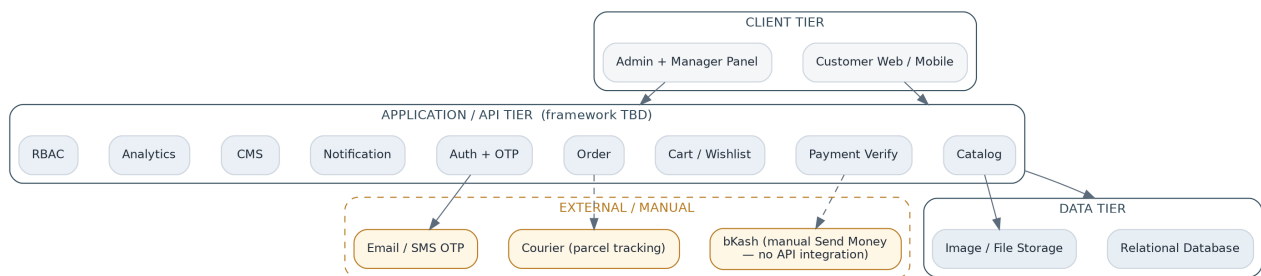


Fig 6. Reference architecture. Final framework pending your stack decision (\$11C).

## 10. Open issues, conflicts & gaps

1. **Order states under-specified** — 3 listed, ≥6 implied (§6).
2. **Category lists disagree** — admin: Winter/Summer + Accessories→Cap→Mask; storefront: Winter/Summer Collection + **Sale** (which is really a discount filter, not a category).
3. **Featured products in two places** — Products module and Website Management. Pick one source of truth.
4. **OTP placement** — registration vs checkout vs both (§7).
5. **Coupons undefined** but used in cart; Reviews / Inventory / Notifications / Reports / Settings / Activity Logs named but unspecified (§3).
6. **Billing address** — optional on storefront, listed plainly in admin checkout. Confirm.
7. **Address granularity mismatch** — admin profile uses Upazila/Union/Division/Postal; storefront address fields looser. Unify.
8. **No refund / return workflow** despite a Return & Refund Policy page. Define or scope out.
9. **Stock vs variant stock** — tracked per product or per size/color variant?
10. **Payment proof optional** — verification weakness (§8).

## 11. Decisions needed before Claude Code builds

**A. Thesis framing.** Is this a *software-engineering build project* (the system is the deliverable) or a *research study* (a question + evaluation)? This determines whether the report/poster needs Abstract / Research Question / Methodology / Evaluation sections.

**B. Build scope.** Build *all* modules, or commit to a defensible **core** (catalogue + cart + checkout + manual bKash + order lifecycle + RBAC + analytics) and label the rest *future work*? A thesis rarely builds 16 admin modules + 16 storefront sections well.

**C. Tech stack.** Name it (e.g. MERN, Laravel + MySQL, Django + Postgres, Next.js + Postgres). This finalises the architecture diagram and the non-functional requirements.

Once A–C are answered and the §10 conflicts resolved, this specification is build-ready.

## 12. Appendix — source typos to fix before any poster / print

canceled → cancelled · orgeder → order · pleased → placed · musk → mask · sandme → same · Foote → Footer · Upojela → Upazila · required → required. Use one spelling of 'cancelled' throughout.